

CHAPTER 1: COMPUTER NETWORK AND COMMUNICATION

1. What does the term "Telecommunication" mean?

- a. Processing data
- b. Transmitting information over a distance
- c. Storing files
- d. Drawing diagrams

2. Which of the following is a guided (wired) transmission medium?

- a. Wi-Fi
- b. Bluetooth
- c. Optical Fiber
- d. Satellite

3. What is the standard length of an IPv6 address?

- a. 32 bits
- b. 64 bits
- c. 128 bits
- d. 256 bits

4. Which connector is primarily used for Ethernet networking cables like CAT6?

- a. USB
- b. RJ-45
- c. HDMI
- d. VGA

5. What does the acronym "PAN" stand for in networking?

- a. Personal Area Network
- b. Public Area Network
- c. Private Access Network
- d. Primary Area Network

6. Which networking device operates by regenerating weak signals over long distances?

- a. Switch
- b. Router
- c. Repeater
- d. Bridge

7. Which protocol is responsible for automatically assigning IP addresses to devices on a network?

- a. SMTP
- b. FTP
- c. HTTP
- d. DHCP

8. In which data communication mode does data flow in both directions, but only one direction at a time?

- a. Simplex
- b. Half Duplex
- c. Full Duplex
- d. Multi Duplex

9. Which unit is commonly used to measure network throughput and bandwidth?

- a.** Hertz (Hz)
- b.** Bits per second (bps)
- c.** Pixels (px)
- d.** Gigabytes (GB)

10. Which network topology uses a single central cable (backbone) to which all nodes are connected?

- a.** Star Topology
- b.** Ring Topology
- c.** Bus Topology
- d.** Hybrid Topology

11. How does a Switch differ from a Hub in a computer network?

- a.** A switch converts optical signals to electrical signals.
- b.** A switch sends data only to the specific intended computer.
- c.** A switch broadcasts data to all connected computers.
- d.** A switch connects completely different types of networks.

12. Why is broadband considered superior to traditional dial-up internet?

- a.** It is entirely wireless.
- b.** It uses older telephone cables.
- c.** It allows sending and receiving large amounts of data at high speeds.
- d.** It only supports simplex communication.

13. What is the primary difference between the Internet and an Intranet?

- a.** The Internet is a private network, whereas an intranet is a global public network.
- b.** The Internet connects globally, whereas an Intranet is a private network for an organization's internal staff.
- c.** The Internet uses cables, whereas an Intranet uses satellites.
- d.** The Internet uses IPv4, whereas an Intranet uses IPv6 exclusively.

14. Why was IPv6 introduced to replace IPv4?

- a.** IPv4 cannot be used for Wi-Fi.
- b.** IPv4 is extremely slow for mobile phones.
- c.** The rapid growth of connected devices led to a shortage of available IPv4 addresses.
- d.** IPv4 addresses are too hard to memorize.

15. Why is data broken down into "data packets" before transmission?

- a.** To increase the cost of transmission.
- b.** To make data transfer faster, more reliable, and efficient.
- c.** To convert text into multimedia.
- d.** To prevent the use of routers.

16. How does data flow in a Ring Topology?

- a.** Through a central hub.
- b.** In a closed loop, sequentially from one computer to another.
- c.** Randomly to any available node.
- d.** Through a central backbone cable directly to the destination.

17. Which statement best describes a Peer-to-Peer network architecture?

- a.** One central server controls all resources.
- b.** It is exclusively used for Wide Area Networks.
- c.** All nodes have equal relationships and can act as both clients and servers.
- d.** It requires complex administration and high-end server hardware.

18. What is the function of a Media Converter in a network?

- a.** To amplify weak Wi-Fi signals.
- b.** To automatically assign IP addresses.
- c.** To convert electrical impulses from copper cables to light signals for fiber optics (and vice versa).
- d.** To connect multiple computers in a star topology.

19. Why is optical fiber widely used by Internet Service Providers (ISPs)?

- a.** It uses radio waves which are cheap.
- b.** It allows data to travel at very high speeds with minimal loss over long distances.
- c.** It is easily bendable and never breaks.
- d.** It is the only cable that works with RJ-45 connectors.

20. How does a router differ from a bridge?

- a.** A router connects similar networks, while a bridge connects to the internet.
- b.** A router forwards data among different networks using IP addresses, while a bridge connects networks with similar protocols.

c. A router generates IP addresses, while a bridge regenerates weak signals.

d. A router is a wireless device, while a bridge is always wired.

21. A school needs to connect 30 computers in a single computer lab.

Which type of network coverage is most appropriate?

a. PAN

b. LAN

c. MAN

d. WAN

22. You want to check if your computer can successfully reach www.google.com. Which command should you type in the command prompt?

a. ipconfig

b. tracert

c. nslookup

d. ping

23. A company wants to share its internal database with trusted external suppliers securely. Which network implementation should they use?

a. Intranet

b. Extranet

c. Internet

d. PAN

24. You are setting up a home network and need to connect your smartphone to your laptop wirelessly to transfer a small file from a few feet away. Which technology should you use?

- a.** Optical Fiber
- b.** Satellite
- c.** Bluetooth
- d.** MAN

25. The signal from your wireless router is too weak to reach the first floor of your house. Which device should you install to solve this problem?

- a.** Switch
- b.** Media Converter
- c.** Repeater (Range Extender)
- d.** Bridge

26. An organization has multiple branches within Kathmandu valley and wants to connect their networks. Which network type is formed?

- a.** PAN
- b.** LAN
- c.** MAN
- d.** WAN

27. A network administrator needs to find the specific IP address and Default Gateway of a Windows computer. Which command should they execute?

- a.** ipconfig
- b.** ping

- c.** tracert
- d.** nslookup

28. A smart card is used to track inventory items automatically in a warehouse using radio waves. Which technology is being applied here?

- a.** Bluetooth
- b.** RFID
- c.** RJ-45
- d.** 5G Cellular

29. You are designing a network that requires central control, high security, and centralized data backups. Which network architecture should you choose?

- a.** Peer-to-Peer
- b.** Client-Server
- c.** Ring Topology
- d.** Unguided Media

30. A television station broadcasts a live football match to millions of viewers, but viewers cannot send data back to the station through the TV. Which communication mode is this?

- a.** Simplex
- b.** Half Duplex
- c.** Full Duplex
- d.** Complex Duplex

31. Which of the following is a guided (wired) transmission medium?

- a.** Optical Fiber

b. Bluetooth

c. Wi-Fi

d. Satellite

32. What is the full form of PAN?

a. Personal access network

b. Private area network

c. Personal area network

d. Public access network

33. What is the main purpose of a repeater?

a. Amplify signals

b. Connect different networks and direct data

c. Store data

d. Connect nodes in a loop

34. Which device strengthens weak network signals for long distances?

a. Switch

b. Router

c. Repeater

d. Bridge

35. How does bandwidth affect internet service?

a. More bandwidth = Slower speed

b. Less bandwidth = Higher speed

c. More bandwidth = Faster speed

d. Bandwidth does not affect speed

36. What is the main purpose of a CAT6 cable in a network?

a. To transmit data wirelessly

b. To connect devices using radio signals

c. To send signals through space using microwaves

d. To provide high-speed data transmission through copper wires

37. Which protocol is commonly used for sending emails?

a. HTTP

b. FTP

c. SMTP

d. DHCP

38. What is the length of an IPv4 address?

a. 16 bit

b. 32 bit

c. 64 bit

d. 128 bit

39. What do you call the technology that allows connecting phones to the internet with laptops without cables?

a. Router

b. Bluetooth

c. Repeater

d. Bridge

40. What is the connection pattern of computers in a network called?

a. Protocol

b. Topology

- c. Twisted pair
- d. Structure

41. What unit is used to measure bandwidth?

- a. Hertz
- b. Bits per second (bps)
- c. Kilometers
- d. Joules

42. What is the main purpose of a firewall in a computer network?

- a. Increase speed
- b. Store data
- c. Protect against unauthorized access
- d. Reduce signal interference

43. Which of the following is a broadband internet connection?

- a. DSL
- b. Fiber optic
- c. Cable internet
- d. All of the above

44. Which topology should be chosen by a company that wants a network that will not fail even if one computer fails, and makes troubleshooting easy?

- a. Bus topology
- b. Ring topology
- c. Star topology
- d. Hybrid topology

45. IPv6 is used because:

- a. IPv4 addresses are running out.
- b. It is faster than IPv4.
- c. It uses fewer bits.
- d. It is only for LAN.

46. Which of the following is a broadband Internet connection?

- a. DSL
- b. Fiber optic
- c. Cable internet
- d. All of the above

47. What is throughput?

- a. Theoretical speed of a network
- b. Actual data transferred in a given time
- c. Length of a network cable
- d. Number of users

48. What is a data packet?

- a. A physical network device
- b. A unit of data sent over a network
- c. A type of wireless method
- d. A security tool

49. Which of the following is a type of bounded (guided) media?

- a. Fiber optic
- b. Infrared
- c. Microwave

d. Laser

50. Which term refers to sending data from Earth to a satellite?

a. Downlink

b. Modulate

c. Uplink

d. Download

51. What is the RJ45 connector mainly used for?

a. USB connections

b. Telephone lines

c. Ethernet networking

d. Fiber optics

52. What is the connection pattern of computers in a network called?

a. Protocol

b. Topology

c. Twisted pair

d. Structure

53. Which topology uses a hub to connect all devices?

a. Ring topology

b. Bus topology

c. Star topology

d. Hybrid topology

54. What type of network connects LANs over large areas?

a. PAN

b. MAN

c. WAN

d. CAN

55. Which of the following are Internet services?

a. IRC

b. Telnet

c. Email

d. All of the above

56. Which protocol is used to transfer files between computers?

a. FAQ

b. IRC

c. FTP

d. TPF

57. What is the length of an IPv4 address?

a. 16 bits

b. 32 bits

c. 64 bits

d. 128 bits

58. What does an IP address identify?

a. A software

b. A network cable

c. A specific device on the network

d. A computer brand

59. Which protocol is commonly used for sending emails?

- a.** HTTP
- b.** FTP
- c.** SMTP
- d.** DHCP

60. Which device strengthens weak network signals for long distances?

- a.** Switch
- b.** Router
- c.** Repeater
- d.** Bridge

61. Which term refers to the maximum data-carrying capacity of a communication channel?

- a.** Bandwidth
- b.** Throughput
- c.** Frequency
- d.** Broadband

62. Why is Bluetooth more suitable than optical fiber for short-range device communication?

- a.** It allows wireless communication between nearby devices.
- b.** It transmits data using light signals.
- c.** It is mainly used for long-distance internet connections.
- d.** It is used only for television broadcasting.

63. A network administrator wants to check whether two computers on a network can communicate successfully. Which command should be used for this purpose?

- a.** ping
- b.** format
- c.** compile
- d.** merge

64. What does LAN stand for?

- a.** Local Area Network
- b.** Long Area Network
- c.** Large Area Network
- d.** Logical Area Network

65. Why is a computer network used?

- a.** To increase the size of a monitor
- b.** To share resources and exchange information
- c.** To convert analog signals into digital signals
- d.** To create databases

66. A school wants to connect computers within a single building. Which network should be used?

- a.** PAN
- b.** LAN
- c.** MAN
- d.** WAN

67. Which communication mode allows data transmission in both directions simultaneously?

- a.** Simplex
- b.** Half Duplex
- c.** Full Duplex
- d.** Multiplex

68. Why is a switch generally preferred over a hub in a network?

- a.** It amplifies signals
- b.** It broadcasts data to all devices
- c.** It sends data only to the intended device
- d.** It converts digital signals into analog signals

69. A company needs to connect branches located in different cities. Which network is suitable?

- a.** PAN
- b.** LAN
- c.** MAN
- d.** WAN

70. What does IP stand for?

- a.** Internet Protocol
- b.** Internal Protocol
- c.** Internet Program
- d.** Internal Program

71. Why is fiber optic cable considered better than twisted pair cable for long-distance communication?

- a.** It uses electrical signals
- b.** It is cheaper
- c.** It provides higher speed and less signal loss
- d.** It does not require maintenance

72. Which transmission medium would you choose for high-speed internet in a city?

- a.** Twisted Pair Cable
- b.** Fiber Optic Cable
- c.** Coaxial Cable
- d.** Telephone Cable

73. How is throughput different from bandwidth?

- a.** Bandwidth is the actual data transfer rate, while throughput is the theoretical maximum.
- b.** Throughput is the actual data transfer rate, while bandwidth is the maximum possible rate.
- c.** Both mean the same in every situation.
- d.** Throughput measures data errors, while bandwidth measures frequency.

74. Which mobile communication technology provides the fastest data transfer and lowest latency among the following?

- a.** 3G
- b.** 4G
- c.** 5G

d. Broadband

75. Which of the following is an example of wired communication media?

a. Wi-Fi

b. Bluetooth

c. Optical fiber

d. Satellite

76. What is the main purpose of CAT6 cable in a network?

a. To transmit data wirelessly

b. To connect devices using radio signals

c. To send signals through space using microwaves

d. To provide high-speed data transmission through copper wires

77. A company needs to connect two offices located 5 kilometers apart with high-speed and interference-free data transmission. Which medium should they choose?

a. Bluetooth

b. Optical fiber cable

c. Wi-Fi

d. RFID

78. How does wireless media differ from wired media?

a. Wireless media always use metal cables for signal transmission.

b. Wired media are slower but more secure than wireless media.

c. Wireless media transmit signals through air or space instead of cables.

d. Wired media cannot connect computers at all.

79. Your computer teacher wants to share files between a smartphone and a laptop quickly without using cables or the Internet. Which technology is most suitable?

a. Satellite communication

b. Bluetooth

c. Optical fiber

d. CAT6 cable

80. What is the primary function of an RJ45 connector?

a. To convert signals between wired and wireless networks

b. To connect fiber optic cables to devices

c. To connect Ethernet cables to network devices such as computers and switches

d. To amplify data signals for long-distance transmission

81. Which statement best describes the function of an RJ45 connector?

a. It amplifies wireless signals.

b. It connects fiber optic cables to network switches.

c. It connects twisted-pair cables to network devices like routers and computers.

d. It converts analog signals into digital form.

82. Why is a media converter important in modern network setups?

a. It increases the number of Wi-Fi connections.

b. It allows different types of network cabling (fiber and copper) to work together efficiently.

c. It stores data packets temporarily.

d. It boosts the power of wireless signals.

83. Which networking device is used to regenerate or amplify signals in a network to extend transmission distance?

a. Hub

b. Repeater

c. Switch

d. Router

84. How does a hub function in a network?

a. It filters traffic between two networks.

b. It sends data only to the destination device.

c. It broadcasts data to all connected devices.

d. It converts analog signals into digital signals.

85. A small office wants to connect multiple computers and ensure that data is sent only to the intended device to reduce unnecessary traffic. Which device should they use?

a. Hub

b. Switch

c. Repeater

d. Bridge

86. What is the main difference between a bridge and a switch?

a. A bridge connects two different networks, while a switch connects devices within the same network.

b. A bridge works only with wireless signals, while a switch works with wired signals.

c. A bridge amplifies signals, while a switch filters them.

d. A bridge stores data permanently, while a switch transmits data temporarily.

87. A network administrator wants to connect two different networks and also manage data traffic between them. Which device should be used?

a. Switch

b. Bridge

c. Repeater

d. Router

88. In which network topology are all devices connected to a single central cable called the backbone?

a. Star topology

b. Ring topology

c. Bus topology

d. Hybrid topology

89. Which of the following best describes a Star topology?

a. Each computer is connected to every other computer directly.

b. All computers are connected to a single central device such as a switch or hub.

c. All computers share one main communication line.

d. Data travels in one direction only around a loop.

90. A company wants a network that will not stop working even if one computer fails, and troubleshooting is easy. Which topology should they choose?

- a.** Bus topology
- b.** Ring topology
- c.** Star topology
- d.** Hybrid topology

91. What is the main disadvantage of using a Ring topology?

- a.** It requires more cable than a star network.
- b.** If one computer fails, it can affect the entire network.
- c.** It cannot connect more than two computers.
- d.** It does not allow data to travel in a loop.

92. Why would a network administrator choose to use a Hybrid topology?

- a.** It is the cheapest topology to set up.
- b.** It combines the advantages of two or more topologies for better performance and reliability.
- c.** It allows data to move in only one direction.
- d.** It is the simplest topology to install.

93. Which network is designed for personal use, around a person or a workspace?

- a.** LAN
- b.** MAN
- c.** WAN
- d.** PAN

94. How does a LAN (Local Area Network) differ from a WAN (Wide Area Network)?

- a.** LAN covers a larger area than WAN.
- b.** WAN connects devices within a single building.
- c.** LAN covers a small area like a home or office while WAN connects networks across cities or countries.
- d.** WAN is faster and cheaper than LAN in all cases.

95. A university with several buildings across the same city wants to connect all its departmental networks. Which type of network should it use?

- a.** PAN
- b.** MAN
- c.** LAN
- d.** WAN

96. Which statement best explains the relationship among PAN, LAN, MAN, and WAN in terms of coverage area?

- a.** All have equal coverage areas.
- b.** $PAN < LAN < MAN < WAN$ in terms of increasing geographical range.
- c.** $LAN < PAN < WAN < MAN$ in terms of increasing size.
- d.** $WAN < MAN < LAN < PAN$ in terms of range.

97. What is the main function of a server in a client-server network?

- a.** To provide data and services to client computers
- b.** To store personal files only
- c.** To connect directly to the Internet

d. To act as a backup power supply

98. A teacher wants to create a small classroom network where students can share files directly from one computer to another without a server. Which network type should be used?

- a.** Client-Server Network
- b.** Peer-to-Peer Network
- c.** Metropolitan Area Network
- d.** Wide Area Network

99. What does an IP address identify in a network?

- a.** The unique address of a device on the network
- b.** The type of cable used for connection
- c.** The speed of Internet connection
- d.** The name of a website

100. Which of the following correctly distinguishes IPv4 from IPv6?

- a.** IPv4 uses 128-bit addresses, while IPv6 uses 32-bit addresses.
- b.** IPv4 uses 32-bit addresses, while IPv6 uses 128-bit addresses.
- c.** Both IPv4 and IPv6 use 64-bit addresses.
- d.** IPv6 is only used in wireless networks.

101. A company is running out of available IP addresses due to rapid device growth. Which version of IP should it adopt to solve this problem, and why?

- a.** IPv4, because it supports simpler addressing
- b.** IPv6, because it provides a much larger address space
- c.** IPv4, because it is faster than IPv6

d. IPv6, because it uses private addresses only

102. Which of the following best defines the Internet?

- a.** A private network used only within an organization
- b.** A global network connecting millions of computers worldwide
- c.** A network limited to employees and selected business partners
- d.** A local area network used for file sharing

103. How does an Intranet differ from an Extranet?

- a.** An Intranet allows access only to employees, while an Extranet also allows access to trusted external users.
- b.** An Extranet is larger than the Internet.
- c.** An Intranet connects multiple companies through satellites.
- d.** Both Intranet and Extranet are open to the general public.

104. A university wants to share study materials and schedules only with its students and staff. Which type of network should it implement?

- a.** Intranet
- b.** Internet
- c.** Extranet
- d.** Metropolitan Area Network

105. Which of the following best describes an intranet?

- a.** A network that connects multiple organizations
- b.** A private network used within an organization
- c.** A public network accessible to anyone
- d.** A network that connects devices over long distances

106. How does an extranet differ from an intranet?

- a.** An extranet is accessible only to internal users, while an intranet is accessible to external users.
- b.** An extranet is a public network, while an intranet is a private network.
- c.** An extranet connects multiple organizations, while an intranet connects devices within a single organization.
- d.** There is no difference between them.

107. What is throughput?

- a.** Theoretical speed of a network
- b.** Actual data transferred in a given time
- c.** Length of a network cable
- d.** Number of users

108. What is a data packet?

- a.** A physical network device
- b.** A unit of data sent over a network
- c.** A type of wireless method
- d.** A security tool

109. Which of the following is a type of bounded (guided) media?

- a.** Fiber optic
- b.** Infrared
- c.** Microwave
- d.** Laser

110. Which term refers to sending data from Earth to a satellite?

- a.** Downlink

b. Modulate

c. Uplink

d. Download

111. What is the connection pattern of computers in a network called?

- a.** Protocol
- b.** Topology
- c.** Twisted pair
- d.** Structure

112. Which topology uses a hub to connect all devices?

- a.** Star topology
- b.** Ring topology
- c.** Bus topology
- d.** Hybrid topology

113. What type of network connects LANs over large areas?

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- b.** MAN
- c.** WAN
- d.** CAN

114. Which of the following are Internet services?

- a.** IRC
- b.** Telnet
- c.** Email
- d.** All of the above

115. Which protocol is used to transfer files between computers?

- a.** FAQ
- b.** IRC
- c.** FTP
- d.** TPF

116. What is the length of an IPv4 address?

- a.** 16 bits
- b.** 32 bits
- c.** 64 bits
- d.** 128 bits

117. What does an IP address identify?

- a.** A software
- b.** A network cable
- c.** A specific device on the network
- d.** A computer brand

118. Which protocol is commonly used for sending emails?

- a.** HTTP
- b.** FTP
- c.** SMTP
- d.** DHCP

119. Which device strengthens weak network signals for long distances?

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- b.** Router
- c.** Repeater

d. Bridge

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b. Telephone lines

c. Ethernet networking

d. Fiber optics

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b. Router

c. Repeater

d. Bridge

CHAPTER 2: DATABASE MANAGEMENT SYSTEM (DBMS)

1. Which of the following best defines a database?
 - a. A standalone application for editing text
 - b. A systematically organized collection of related data
 - c. A hardware device used to store internet files
 - d. A programming language used for web development
2. What does DBMS stand for?
 - a. Data Base Management Software
 - b. Digital Base Management Software
 - c. Database Management System
 - d. Data Backup Management System
3. What term is used to describe a collection of raw, unorganized facts and figures before they are processed?
 - a. Information
 - b. Reports
 - c. Data
 - d. Records
4. Which of the following belongs to the Data Definition Language (DDL) category?
 - a. SELECT
 - b. INSERT
 - c. CREATE
 - d. UPDATE
5. A row in a database table is also called a:
 - a. Record
 - b. Field
 - c. Column
 - d. Report
6. Which SQL data type is specifically designed to store text of variable lengths?
 - a. INT
 - b. VARCHAR
 - c. DATE
 - d. BOOLEAN
7. Which type of key is specifically designed to ensure every record in a table is uniquely identified?
 - a. Foreign Key
 - b. Primary Key
 - c. Alternate Key
 - d. Candidate Key
8. Which wildcard character is used to represent any sequence of zero or more characters in SQL?
 - a. _ (underscore)
 - b. * (asterisk)
 - c. % (percent sign)
 - d. ? (question mark)

9. If a user needs to fetch and display specific records from a table, which SQL command must they use?

- a.** SELECT
- b.** RETRIEVE
- c.** EXTRACT
- d.** DISPLAY

10. Which SQL command is used to permanently delete an entire table and its structure from a database?

- a.** DELETE
- b.** REMOVE
- c.** DROP
- d.** TRUNCATE

11. A hospital records patient temperatures and symptoms, processes them through a system, and generates a structured health report. What does the final health report represent?

- a.** Raw Data
- b.** Meaningful Information
- c.** A Database Schema
- d.** An SQL Query

12. In a banking database, the Loans table contains a column named CustomerID which connects each loan to a specific person in the Customers table. What is CustomerID in the Loans table?

- a.** Primary Key
- b.** Composite Key

- c.** Foreign Key
- d.** Alternate Key

13. A retail manager needs to find all products that have the word "Phone" anywhere in their product name. Which SQL clause is required?

- a.** WHERE ProductName = 'Phone'
- b.** WHERE ProductName LIKE 'Phone'
- c.** WHERE ProductName LIKE '%Phone%'
- d.** WHERE ProductName LIKE '_Phone'

14. A school wants to record the exact date and time a student submitted an online assignment. Which data type should the database designer apply to this column?

- a.** DATE
- b.** DATETIME
- c.** TIME
- d.** VARCHAR

15. An HR manager notices that an employee's salary is listed incorrectly and needs to fix it. Which SQL command must they execute?

- a.** ALTER
- b.** MODIFY
- c.** UPDATE
- d.** CHANGE

16. A developer needs to add a new column named BloodGroup to an existing Patients table. Which SQL command should be used?

- a.** CREATE COLUMN BloodGroup IN Patients;

- b.** INSERT INTO Patients ADD BloodGroup;
- c.** ALTER TABLE Patients ADD BloodGroup VARCHAR(5);
- d.** UPDATE Patients SET Column = 'BloodGroup';

17. An inventory system runs the following query: SELECT * FROM Inventory;. What will be the output?

- a.** It will delete all inventory records.
- b.** It will display all columns and all records currently in the Inventory table.
- c.** It will create a new table named Inventory.
- d.** It will only display the primary keys of the Inventory table.

18. A business wants to automatically assign the value 'Active' to an employee's status if the HR department doesn't specify a status during data entry. Which constraint should be applied?

- a.** UNIQUE
- b.** DEFAULT
- c.** CHECK
- d.** NOT NULL

19. If you have a table named Temporary_Data and you want to completely erase the table so its structure no longer exists in the database, which query should you run?

- a.** DELETE * FROM Temporary_Data;
- b.** REMOVE TABLE Temporary_Data;
- c.** DROP TABLE Temporary_Data;
- d.** TRUNCATE TABLE Temporary_Data;

20. To insert a new customer's details into the Customers table, which SQL syntax format is correct?

- a.** CREATE Customers VALUES (...);
- b.** UPDATE Customers SET (...);
- c.** SELECT INTO Customers VALUES (...);
- d.** INSERT INTO Customers VALUES (...);

21. What's the primary purpose of a database?

- a.** Playing games
- b.** Organizing and storing data
- c.** Watching videos
- d.** Drawing pictures

22. Which of the following is processed data?

- a.** Raw marks
- b.** Student result sheet
- c.** Roll numbers
- d.** Attendance figures

23. Which software is an example of a DBMS?

- a.** MS Paint
- b.** MySQL
- c.** VLC Player
- d.** Notepad

24. Which data type is suitable for storing age?

- a.** VARCHAR
- b.** INT

c. DATE

d. TEXT

25. Which data type is best for storing a person's name?

a. INT

b. VARCHAR

c. FLOAT

d. DATE

26. A row in a table is called a:

a. Field

b. Record

c. VARCHAR

d. DATE

27. What does a column represent in a table?

a. Record

b. Database

c. Field

d. Report

28. Which key uniquely identifies a record?

a. Foreign Key

b. Composite Key

c. Primary Key

d. Alternate Key

29. A foreign key is mainly used to:

a. Create relationships between tables

b. Store passwords

c. Delete records

d. Create reports

30. Which object is used to retrieve information from tables?

a. Query

b. Record

c. Field

d. Key

31. Which SQL command creates a database?

a. INSERT

b. CREATE

c. DELETE

d. UPDATE

32. Which command modifies the structure of a table?

a. ALTER

b. INSERT

c. SELECT

d. DELETE

33. Which command is used to modify data in a table?

a. UPDATE

b. INSERT

c. SELECT

d. ALTER

34. Which command displays data from a table?

a. UPDATE

b. INSERT

c. SELECT

d. ALTER

35. What is the primary purpose of a foreign key?

a. To store only text data

b. To uniquely identify each record in its own table

c. To establish and enforce relationships between tables

d. To combine multiple fields into one

36. Which SQL command changes existing records?

a. UPDATE

b. CREATE

c. DROP

d. SELECT

37. In a one-to-many relationship between teachers and classes, which statement is true?

a. Each teacher teaches only one class, and each class has only one teacher.

b. Each teacher can teach multiple classes, and each class has only one teacher.

c. Each teacher teaches only one class, and each class can have multiple teachers.

d. Each teacher can teach multiple classes, and each class can have multiple teachers.

38. Which clause filters records according to a condition?

a. VALUES

b. WHERE

c. INTO

d. FROM

39. In SQL, the '%' wildcard is used with which clause?

a. SELECT

b. WHERE

c. LIKE

d. INSERT

40. Which report shows complete details of records?

a. Summary Report

b. Detail Report

c. Aggregate Report

d. Filtered Report

41. Which of the following best describes a Database?

a. A collection of unrelated data

b. A collection of interrelated data

c. A single table of data

d. A set of random files

42. What does DBMS stand for?

- a.** Database Management System
- b.** Digital Base Management Software
- c.** Data Basic Managing System
- d.** DataBank Management System

43. Which language is used to define the structure of a database (create/alter/drop tables)?

- a.** DDL
- b.** DML
- c.** DCL
- d.** TCL

44. Which of the following is a DML command?

- a.** CREATE
- b.** ALTER
- c.** INSERT
- d.** DROP

45. In DBMS, a table is also known as a:

- a.** Tuple
- b.** Relation
- c.** Attribute
- d.** Field

46. What does SQL stand for?

- a.** Sequential Query Language
- b.** Structured Query Language

c. Server Query Language

d. Simple Query Language

47. Which of the following is used to uniquely identify a record in a table?

- a.** Foreign Key
- b.** Primary Key
- c.** Candidate Key
- d.** Alternate Key

48. Which of the following is an advantage of DBMS?

- a.** Data Redundancy
- b.** Data Inconsistency
- c.** Reduced Data Redundancy
- d.** Data Duplication

49. The rows in a table are also called:

- a.** Tuples/Records
- b.** Fields
- c.** Entities
- d.** Columns

50. The columns in a table are also called:

- a.** Tuples
- b.** Attributes
- c.** Entities
- d.** Records

51. Which key is used to link two tables?

- a. Primary Key**
- b. Foreign Key**
- c. Super Key**
- d. Composite Key**

52. A Primary Key column:

- a. Can have duplicate values**
- b. Can have NULL values**
- c. Can have only text values**
- d. Cannot be NULL**

53. What is a Database?

- a. Collection of programs**
- b. Collection of unrelated data**
- c. Collection of interrelated data**
- d. Collection of files only**

54. Which of the following is a DML command?

- a. CREATE**
- b. ALTER**
- c. INSERT**
- d. DROP**

55. Which key uniquely identifies a record in a table?

- a. Foreign Key**
- b. Primary Key**
- c. Super Key**

d. Alternate Key

56. Which key refers to the primary key of another table?

- a. Primary Key**
- b. Foreign Key**
- c. Alternate Key**
- d. Candidate Key**

57. Which command is used to remove a table from a database?

- a. DELETE**
- b. REMOVE**
- c. DROP**
- d. TRUNCATE**

58. Which command is used to change existing data in a table?

- a. ALTER**
- b. MODIFY**
- c. UPDATE**
- d. SELECT**

59. Which command is used to retrieve data from a database?

- a. GET**
- b. SELECT**
- c. RETRIEVE**
- d. OUTPUT**

60. Which SQL statement is used to insert a new record?

- a. ADD**

- b. INSERT**
- c. UPDATE**
- d. CREATE**

61. Which of the following describes what a query does?

- a. Stores data permanently**
- b. Retrieves or manipulates data based on a condition**
- c. Deletes the database**
- d. Formats the report**

62. Which SQL command is used to delete a database?

- a. REMOVE DATABASE**
- b. DELETE DATABASE**
- c. DROP DATABASE**
- d. CLEAR DATABASE**

63. The SQL command used to delete all rows in a table (keeping structure) is:

- a. Data Manipulation Language**
- b. DELETE**
- c. TRUNCATE**
- d. REMOVE**

64. Which type of key uniquely identifies each record in a table?

- a. Foreign Key**
- b. Composite Key**
- c. Primary Key**
- d. Candidate Key**

65. What is the primary purpose of a foreign key?

- a. To store only text data.**
- b. To uniquely identify each record in its own table.**
- c. To establish and enforce relationships between tables.**
- d. To combine multiple fields into one.**

66. In a one-to-many relationship between teachers and classes, which statement is true?

- a. Each teacher teaches only one class, and each class has only one teacher.**
- b. Each teacher can teach multiple classes, and each class has only one teacher.**
- c. Each teacher teaches only one class, and each class can have multiple teachers.**
- d. Each teacher can teach multiple classes, and each class can have multiple teachers.**

67. Which SQL command is used to add new rows to a table?

- a. SELECT**
- b. UPDATE**
- c. INSERT**
- d. DELETE**

68. Which SQL command is used to modify existing data in a table?

- a. SELECT**
- b. UPDATE**
- c. INSERT**

d. DELETE

69. Which SQL clause is used to specify conditions for data retrieval?

a. SELECT

b. FROM

c. WHERE

d. ORDER BY

70. What does the SQL LIKE clause help you do?

a. Perform exact matches.

b. Search for patterns in data.

c. Sort the data.

d. Define data types.

71. Which SQL data type is suitable for storing names and addresses, allowing for variable length?

a. CHAR

b. TEXT

c. INT

d. VARCHAR

72. What is the primary difference between CHAR(n) and VARCHAR(n) data types?

a. CHAR stores only numeric data, while VARCHAR stores text.

b. CHAR has a fixed length, while VARCHAR has a variable length up to n characters.

c. VARCHAR is used for dates, while CHAR is for text.

d. There is no significant difference between them.

73. Which key uniquely identifies each record in a database table?

a. Composite key

b. Primary key

c. Candidate key

d. Unique key

74. You want to see all the student information stored in a database table.

Which SQL command should you use to view the data?

a. INSERT

b. UPDATE

c. SELECT

d. ALTER

75. What is a database?

a. A collection of programs

b. A collection of related data

c. A computer network

d. An operating system

76. A school wants to store students' details systematically. Which software should be used?

a. Paint

b. MS Access

c. Calculator

d. Media Player

77. A row in a database table is called:

a. Field

- b. Record**
- c. Query**
- d. Report**

78. While creating a student table, which field is most suitable as a primary key?

- a. Name**
- b. Address**
- c. Roll Number**
- d. Section**

79. Which database object is used to retrieve specific data?

- a. Form**
- b. Report**
- c. Query**
- d. Table**

80. A library wants to store book records and member records in separate tables. Which feature helps connect them?

- a. Formatting**
- b. Relationship**
- c. Sorting**
- d. Printing**

81. Which of the following is NOT a function of a DBMS?

- a. Data storage**
- b. Data retrieval**
- c. Word processing**

- d. Data manipulation**

82. Which type of key uniquely identifies each record in a table?

- a. Foreign Key**
- b. Composite Key**
- c. Primary Key**
- d. Candidate Key**

83. What is the primary purpose of a foreign key?

- a. To store only text data.**
- b. To uniquely identify each record in its own table.**
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- b. UPDATE**

- c. INSERT
- d. DELETE

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- b. Search for patterns in data.
- c. Sort the data.
- d. Define data types.

89. Which SQL data type is used for storing only date values (e.g., YYYY-MM-DD)?

- a. TIME
- b. DATETIME
- c. TIMESTAMP
- d. DATE

90. Which SQL data type is suitable for storing names and addresses, allowing for variable length?

- a. CHAR
- b. TEXT
- c. INT
- d. VARCHAR

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- c. VARCHAR is used for dates, while CHAR is for text.
- d. There is no significant difference between them.

CHAPTER 3: MULTIMEDIA

1. What does the term "Multimedia" literally mean?

- a. Single form of media
- b. Multiple forms of media
- c. Manipulated media
- d. Machine media

2. Which of the following is the smallest unit of a raster graphic?

- a. Frame
- b. Vector
- c. Pixel
- d. Curve

3. Which text file format is native to Microsoft Word?

- a. TXT
- b. RTF
- c. DOCX
- d. PDF

4. Which image format is specifically known for supporting simple animations?

- a. JPG
- b. SVG
- c. PNG
- d. GIF

5. What does the "Amplitude" in an audio waveform graph represent?

- a. The length of the audio
- b. The file size of the audio
- c. The loudness of the sound
- d. The pitch of the sound

6. Which unit is used to measure the frame rate of a video?

- a. Hertz (Hz)
- b. Pixels (px)
- c. Frames per second (FPS)
- d. Megabytes (MB)

7. What is the standard pixel dimension for a Full High Definition (FHD) video?

- a. 720 x 480
- b. 1280 x 720
- c. 1920 x 1080
- d. 3840 x 2160

8. Which software is mentioned as a tool for editing graphics?

- a. Audiomass
- b. CapCut
- c. Pixlr X
- d. Microsoft Excel

9. In 3D animation, which dimension is added that is not present in 2D animation?

- a. Width

- b. Height**
- c. Time**
- d. Depth**

10. What is the primary function of the "Cutout" tool in Pixlr X?

- a.** To draw straight lines
- b.** To remove specific portions or backgrounds of an image
- c.** To add text over an image
- d.** To change the overall image format

11. A student needs to insert a company logo onto a website with a dark blue background. The logo itself has no background. Which format should they use?

- a.** JPG
- b.** GIF
- c.** PNG
- d.** MP4

12. A podcast creator wants to record a voiceover in Audiomass and completely remove an awkward 5-second silence in the middle. Which tool should they use?

- a.** Zoom
- b.** Cut (Scissors Icon)
- c.** Export/Download
- d.** Change Layer

13. You are designing a massive outdoor billboard. The graphic needs to be scaled up significantly without becoming blurry or pixelated. Which format is appropriate?

- a.** JPG
- b.** Vector (SVG)
- c.** GIF
- d.** MP4

14. A YouTuber wants to make a seamless visual jump from one video clip to another in CapCut. Which feature should they apply between the two clips?

- a.** A Transition
- b.** An Audio Fade
- c.** A Text Overlay
- d.** A Frame Rate reduction

15. You want to adjust the volume and add an "echo" to a sound file. In Audiomass, under which menu would you find the tool for the echo?

- a.** Trim
- b.** Volume
- c.** Effects (e.g., Reverb)
- d.** Export

16. A user is editing an image in Pixlr X and wants the text to appear perfectly on top of a circular shape. Which panel must they use to rearrange the order of these elements?

- a.** Navigation panel

- b.** Layers panel
- c.** Cutout panel
- d.** Export panel

17. You are downloading a movie to watch on a small, older smartphone with limited storage. Which video resolution would be the most practical choice?

- a.** 4K UHD
- b.** FHD (1920x1080)
- c.** HD (1280x720)
- d.** SD (720x480)

18. A student exports an audio file, but it's too large to send via email. What should they do in Audiomass to reduce the file size before exporting again?

- a.** Export it as an uncompressed WAV file.
- b.** Export it as an MP3 and lower the bitrate/quality.
- c.** Add a Reverb effect to compress the sound.
- d.** Change the audio to a JPG image.

19. To trim the duration of a video clip in the CapCut timeline, what specific action should the user take?

- a.** Click the Export button.
- b.** Drag the handles at the beginning or end of the video clip.
- c.** Change the project's aspect ratio.
- d.** Double-click the preview window.

20. You want to add a speech bubble (a shape) to a photo in Pixlr X. Which tool menu contains the option to insert a new shape?

- a.** Cutout Tool
- b.** Arrange Tool
- c.** Text Tool
- d.** Export Tool

21. Which is a common video file format for online streaming?

- a.** MP4
- b.** JPG
- c.** PNG
- d.** SVG

22. Which multimedia element helps explain ideas using visual elements like charts and diagrams?

- a.** Animation
- b.** Graphics
- c.** Audio
- d.** Text

23. Multimedia is the combination of:

- a.** Text only
- b.** Image and sound only
- c.** Different types of media like text, images, audio, video, and animation
- d.** Hardware and software only

24. Which is a drawback of multimedia according to the provided content?

- a.** Costly and inaccessible hardware/software

- b.** Capture audience attention effectively
- c.** Enhancing personal communication
- d.** Creating immersive experiences

25. Which file format is commonly used for sharing documents online and contains both raster and vector graphics?

- a.** JPG
- b.** PDF
- c.** SVG
- d.** GIF

26. Which is easier to create?

- a.** 2D animation
- b.** 3D animation
- c.** Both are equally difficult
- d.** None of the above

27. Which audio file format is known for its effective compression capabilities and is commonly used for sharing audio files?

- a.** WAV
- b.** AAC
- c.** MP3
- d.** FLAC

28. In raster graphics, images are created with:

- a.** Lines
- b.** Pixels
- c.** Curves

- d.** Both (a) and (b)

29. Which video format is commonly used for storing videos with high compression capabilities?

- a.** AVI
- b.** MP4
- c.** MOV
- d.** MKV

30. Which of the following is a main purpose of multimedia?

- a.** To make information interactive and engaging
- b.** To store data in the computer
- c.** To improve processor speed
- d.** To increase file size

31. Which of the following is a standard text file format that only stores text content?

- a.** RTF
- b.** TXT
- c.** DOCX
- d.** PDF

32. Multimedia is commonly used in:

- a.** Education
- b.** Entertainment
- c.** Business
- d.** All of the above

33. Which term is used to describe characters that we see on the screen and is also known as alpha-numeric values?

- a.** Multimedia
- b.** Notepad
- c.** Text
- d.** RTF

34. PDF files are mainly used to:

- a.** Store images only
- b.** Keep documents the same on all devices
- c.** Play videos
- d.** Edit graphics

35. What type of computer graphics is made up of Bezier curves and maintains quality regardless of its size?

- a.** Raster graphics
- b.** Vector graphics
- c.** PDF
- d.** GIF

36. Which file format is commonly used for sharing documents online and can contain both raster and vector graphics?

- a.** JPG
- b.** PDF
- c.** SVG
- d.** GIF

37. Which of the following is a lossless image format?

- a.** JPG
- b.** PNG
- c.** MP3
- d.** MP4

38. In 3D animation, what are the three dimensions that contribute to the perspective of the animations?

- a.** Width, height, and depth
- b.** Width, length, and size
- c.** Height, ruler, and breadth
- d.** Length and width

39. Which video format is commonly used for storing videos with high compression capabilities?

- a.** AVI
- b.** MP4
- c.** MOV
- d.** MKV

40. What does Resolution in video refer to?

- a.** The number of frames per second.
- b.** The amount of audio compression in a video.
- c.** The number of pixels in an image, expressed as width and height.
- d.** The series of frames played in a video.

41. Which of the following is a raster graphics format?

- a.** SVG

- b. GIF**
 - c. EPS**
 - d. PDF**
- 42. What is the main disadvantage of lossy compression?**
- a. Larger file size**
 - b. No data loss**
 - c. Loss of some original data and quality**
 - d. Can't be used for images**
- 43. Which is a lossless image format?**
- a. JPG**
 - b. PNG**
 - c. MP3**
 - d. MP4**
- 44. What does animation in 3D primarily add over 2D animation?**
- a. Animation in 3D environments (depth)**
 - b. Animation with two dimensions (height and width)**
 - c. Animation using video only**
 - d. No difference**
- 45. Which audio format is uncompressed?**
- a. MP3**
 - b. WAV**
 - c. AAC**
 - d. OGG**
- 46. What is the main advantage of vector graphics over raster graphics?**
- a. Cannot be scaled**
 - b. Requires more memory**
 - c. Can be scaled without loss of quality**
 - d. Supports only black and white**
- 47. Which of the following is used for video compression?**
- a. MPEG**
 - b. JPG**
 - c. GIF**
 - d. WAV**
- 48. Which software is commonly used for creating multimedia presentations?**
- a. Photoshop**
 - b. PowerPoint**
 - c. AutoCAD**
 - d. Excel**
- 49. What is the typical frame rate for smooth video playback?**
- a. 5 fps**
 - b. 12 fps**
 - c. 24-30 fps**
 - d. 60 fps minimum**
- 50. Which graphics component uses bitmaps?**
- a. Vector graphics**
 - b. Raster graphics**

c. Text graphics

d. Line graphics

51. What does multimedia integration improve?

a. Data storage only

b. Engagement and interaction

c. Processor speed only

d. None of the above

52. Which component of multimedia provides interactivity?

a. Audio

b. Video

c. User interface and software

d. Graphics

53. Which file format is most suitable for storing high-quality photographs?

a. JPEG

b. GIF

c. SVG

d. BMP

54. What is the main advantage of lossless compression?

a. Reduces file size with no loss of quality

b. Significantly reduces file size but quality is lost

c. Converts video to audio

d. Makes files larger

55. Which multimedia component is responsible for producing sound?

a. Video

b. Audio

c. Animation

d. Graphics

56. Which multimedia component is primarily used to show visual motion?

a. Text

b. Animation

c. Audio

d. Data

57. Which is a drawback of multimedia according to the content?

a. Costly and inaccessible hardware/software

b. Capturing audience attention effectively

c. Enhancing personal communication

d. Creating immersive experiences

58. Which of the following is a standard text file format that only stores text content?

a. RTF

b. TXT

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a. Width, height, and depth

b. Width, length, and size

c. Height, ruler, and breadth

d. Length and width

65. What is the primary characteristic of 2D animations mentioned in the content?

a. They are modeled and rigged.

b. They are created with three dimensions.

c. They have width, height, and depth.

d. They are drawn frame by frame in animation software.

66. Which type of computer graphic is made up of pixels and loses quality when scaled up?

a. Raster Graphics

b. Vector Graphics

c. PDF

d. GIF

67. Which multimedia file format is best for storing vector graphics?

a. JPG

b. PNG

c. SVG

d. MP3

68. Which of the following is a standard text file format that only stores text content?

- a.** RTF
- b.** TXT
- c.** DOCX
- d.** PDF

69. What does the term "Multimedia" refer to?

- a.** Single type of media
- b.** Many types of media
- c.** A specific type of technology
- d.** Media manipulation

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- a.** Costly and inaccessible hardware/software
- b.** Capturing audience attention effectively
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- a.** They are modeled and rigged.
- b.** They are created with three dimensions.
- c.** They have width, height, and depth.
- d.** They are drawn frame by frame in animation software.

81. Which file format is widely used for lossless image compression on the web?

- a.** JPG
- b.** PNG
- c.** MP3
- d.** MOV

82. You want to save a recording of your voice on the computer. Which of the following file formats should you choose to store the audio?

- a.** PNG
- b.** WAV
- c.** GIF
- d.** PDF

83. What is multimedia?

- a.** Use of only text
- b.** Use of only graphics
- c.** Combination of different media elements (text, graphics, audio, video, animation)
- d.** A single chart

84. A teacher wants to explain the water cycle using pictures, narration, and animation. What technology should be used?

- a.** Database
- b.** Multimedia
- c.** Networking
- d.** Programming

85. Which multimedia element is used to convey written information?

- a.** Audio
- b.** Text
- c.** Animation
- d.** Video

86. A student wants to record a speech for a presentation. Which device is needed?

- a.** Scanner
- b.** Microphone
- c.** Printer
- d.** Plotter

87. Which multimedia component is used to present spoken information?

- a.** Graphics
- b.** Text
- c.** Audio
- d.** Animation

88. Which multimedia element would be most suitable for demonstrating the operation of a machine?

- a.** Video/Animation
- b.** Table
- c.** Table
- d.** Spreadsheet

CHAPTER 4: PROGRAMMING IN PYTHON

1. Consider: `def calculate(a, b=5): return a + b` - what is the output of `print(calculate(10))`?

- a. 50
- b. 15
- c. Syntax Error
- d. An error, since b is required

2. Consider the list `fruits = ["Apple", "Mango", "Banana"]`. If a loop runs for `x in fruits: print(x)`, what happens?

- a. It prints numbers 0, 1, 2.
- b. It prints each fruit name on a new line.
- c. It crashes because `range()` is missing.
- d. It prints the entire list once.

3. What is the critical logic flaw in the following while loop? `while count < 5: print("Running")`

- a. It will not print anything.
- b. `count` must be a string.
- c. It is an infinite loop because `count` is never incremented.
- d. while loops cannot be used with integers.

4. A programmer defines `def sum(a, b): return a+b`. Later, they write `sum(b=5, a=10)`. What argument passing is demonstrated here?

- a. Default arguments
- b. Positional arguments
- c. Keyword (named) arguments

d. Variable-length arguments

5. Evaluate the use of `try...except ZeroDivisionError`. If code placed in the `try` block divides a number by a valid non-zero integer, what does Python do with the `except` block?

- a. It executes the `except` block just in case.
- b. It skips the `except` block entirely and continues normal execution.
- c. It triggers a syntax error.
- d. It deletes the variable to save memory.

6. Why might a data scientist choose the Plotly library over Matplotlib for a web-based dashboard?

- a. Plotly is the only library that can draw bar charts.
- b. Matplotlib cannot read CSV files.
- c. Plotly creates dynamic, interactive charts (zoom, hover tooltips) whereas Matplotlib usually generates static images.
- d. Plotly does not require Python.

7. A user is manipulating a file: `file = open('data.txt', 'r')`, then immediately calls `file.write("Hello")`. What is the result?

- a. "Hello" is appended to the file.
- b. "Hello" overwrites the file.
- c. An `UnsupportedOperation` error occurs because the file is opened in read-only mode.
- d. A new file named `Hello.txt` is created.

8. Analyze the Turtle code snippet: `for i in range(3): turtle.forward(100)
turtle.left(120)`. What shape does this specific algorithm draw?

- a.** Square
- b.** Hexagon
- c.** Circle
- d.** Equilateral Triangle

9. What is the fundamental difference between using `import math` versus `from math import *` in a Python script?

- a.** `import math` requires calling functions as `math.sqrt()`, while `from math import *` imports all functions into the global namespace, allowing just `sqrt()`.
- b.** `from math import *` is strictly for the Turtle library.
- c.** `import math` only imports addition, while the asterisk imports everything.
- d.** There is absolutely no difference in syntax usage.

10. How does the `df[df['Salary'] > 50000]` operation fundamentally function in Pandas?

- a.** It deletes all employees making less than 50000.
- b.** It performs a boolean filtering operation, returning a new DataFrame containing only rows where the 'Salary' column condition is True.
- c.** It adds 50000 to everyone's salary.
- d.** It throws an error if the salaries are stored as integers.

11. Which keyword is used to define a function in Python?

- a.** define

b. function

c. def

d. return

12. A student has written the following Python code: `num = input("Enter a number: ") print(num + 5)` When running the program, an error occurs.

What is the main reason for the error?

- a.** `input()` returns an integer.
- b.** `num` is a string and cannot be added directly to an integer.
- c.** `print()` is used incorrectly.
- d.** The variable is not declared.

13. What is the full form of CSV?

- a.** Common Separated Value
- b.** Computer Stored Value
- c.** Comma Separated Value
- d.** Column Standard Value

14. Study the following Python code: `age = int(input("Enter age: ")) if age >= 18: print("Eligible")` What will happen if the user enters "abc"?

- a.** "Eligible" will be printed.
- b.** The program will stop silently.
- c.** A `ValueError` occurs.
- d.** A `TypeError` occurs.

15. Which operator is used to find the remainder in Python?

- a.** /
- b.** %

c. //

d. **

16. A student writes the following code to check for an even number: num = 7 if num / 2 == 0: print("Even") Why is this code incorrect?

a. % should be used instead of /.

b. num cannot be an integer.

c. if cannot be used.

d. print() should be removed.

17. Which argument allows calling a function using parameter names?

a. Positional argument

b. Keyword argument

c. Default argument

d. Local argument

18. Study the following Python function: def add(a, b): return a+b print(add(5)) What is the problem in the program?

a. return cannot be used.

b. One argument is missing.

c. The function name is incorrect.

d. print() is not necessary.

19. Variables declared inside a function have:

a. Global scope

b. Local scope

c. Unlimited scope

d. Public scope

20. Consider the following list in Python: marks = [45, 67, 89]

print(marks[3]) What happens when the program is executed?

a. 89 is printed.

b. 67 is printed.

c. An IndexError occurs.

d. A SyntaxError occurs.

21. Which loop is mostly used when the number of repetitions is known?

a. while loop

b. do-while loop

c. for loop

d. nested loop

22. Study the following code: import random print(random.randint(1,5))

What does this code do?

a. Generates a random integer from 1 to 5.

b. Generates a decimal number from 1.0 to 5.0.

c. Always prints 5.

d. Always prints 1.

23. Which function generates a random integer?

a. random.choice()

b. random.randint()

c. math.sqrt()

d. pandas.read_csv()

24. A student wants to draw a square using Turtle: for i in range(4):
t.forward(100) t.right(80) Why does the program fail to draw a square?

- a.** range(4) is incorrect.
- b.** right(90) should be used instead of right(80).
- c.** forward() is incorrect.
- d.** A loop is not required.

25. Which library is used for graphical drawing?

- a.** Pandas
- b.** Turtle
- c.** CSV
- d.** Random

26. A student wants to prevent errors while performing division. Which structure is most appropriate?

- a.** if-else
- b.** for loop
- c.** try-except
- d.** function

27. Which library is mainly used for data analysis in Python?

- a.** Turtle
- b.** Math
- c.** Pandas
- d.** Random

28. Which relational operator checks equality?

- a.** !=

b. =

c. ==

d. <=

29. Which statement is used when multiple conditions are checked one after another?

- a.** if
- b.** if-else
- c.** if-elif-else
- d.** nested if

30. Which Pandas function is used to read CSV files?

- a.** pd.write_csv()
- b.** pd.DataFrame()
- c.** pd.read_csv()
- d.** pd.open_csv()

31. In Python, which symbol is used to start a comment?

- a.** //
- b.** #
- c.** /*
- d.** <!--

32. What is the correct file extension for Python files?

- a.** .txt
- b.** .java
- c.** .pyt
- d.** .py

33. Which of the following is a valid variable name?

- a.** 2name
- b.** my-name
- c.** my_name
- d.** myname

34. The function used to take user input in Python is:

- a.** print()
- b.** input()
- c.** scan()
- d.** get()

35. What will be the output of `print(10 // 3)`?

- a.** 3.33
- b.** 3
- c.** 4
- d.** 0

36. Which keyword is used to define a function in Python?

- a.** func
- b.** def
- c.** define
- d.** function

37. What will return do inside a function?

- a.** Stop the program
- b.** Print output
- c.** Send value back

d. Repeat the code

38. Local variables can be used:

- a.** Anywhere in the program
- b.** Only inside the function
- c.** Only outside
- d.** Only in loops

39. Which command imports the math library?

- a.** use math
- b.** include math
- c.** import math
- d.** math.load()

40. The random library is used for:

- a.** Sorting
- b.** Data input
- c.** Graphics
- d.** Random numbers

41. In Turtle graphics, `t.forward(100)` means:

- a.** Draws a circle
- b.** Moves 100 pixels forward
- c.** Turns right
- d.** Erases line

42. Which function hides the turtle cursor?

- a.** `t.stop()`

- b. t.done()**
- c. t.hideturtle()**
- d. t.remove()**

43. Which block is used to handle runtime errors?

- a. try-except**
- b. if-else**
- c. for-in**
- d. def-return**

44. Which of the following is a runtime error (exception)?

- a. Missing colon**
- b. 10/0 (ZeroDivisionError)**
- c. Wrong indentation**
- d. Misspelled keyword**

45. The default mode for the open() function is:

- a. write**
- b. binary**
- c. append**
- d. read**

46. What does mode "a" do in file handling?

- a. Overwrites the file**
- b. Reads the file**
- c. Adds data at the end**
- d. Deletes the file**

47. CSV stands for:

- a. Comma Separated Values**
- b. Common Stored Values**
- c. Computer Save Version**
- d. Combined Sheet Values**

48. Which function reads a CSV file using pandas?

- a. pd.loadcsv()**
- b. pd.read_csv()**
- c. pd.get_csv()**
- d. read.pandas()**

49. Which Matplotlib function displays the chart?

- a. plt.display()**
- b. plt.open()**
- c. plt.graph()**
- d. plt.show()**

50. Which chart type is best for showing parts of a whole?

- a. Line chart**
- b. Pie chart**
- c. Bar chart**
- d. Scatter plot**

51. What is the primary goal of data visualization?

- a. To store data efficiently.**
- b. To perform complex calculations on data.**
- c. To understand data through visual context.**

d. To secure data from unauthorized access.

52. Which of the following is a popular Python library for creating basic graphs like line charts and bar charts?

- a.** Seaborn
- b.** Plotly
- c.** Matplotlib
- d.** Pandas

53. Which type of plot uses dots to represent relationships between variables?

- a.** Bar chart
- b.** Line chart
- c.** Scatter plot
- d.** Pie plot

54. What type of chart uses rectangular bars to represent data categories?

- a.** Scatter plot
- b.** Line chart
- c.** Bar chart
- d.** Pie plot

55. Which data visualization library in Python is built on Matplotlib and offers more advanced statistical visualizations?

- a.** Plotly
- b.** Pandas
- c.** Seaborn
- d.** GGPlot

56. Which Plotly method is used to create a scatter plot?

- a.** scatter()
- b.** line()
- c.** bar()
- d.** pie()

57. Which Plotly Express function is used to create a line chart?

- a.** px.scatter()
- b.** px.line()
- c.** px.bar()
- d.** px.pie()

58. Which Matplotlib function is commonly used to create a pie chart?

- a.** plt.scatter()
- b.** plt.plot()
- c.** plt.bar()
- d.** plt.pie()

59. Which Plotly Express function is used to create a bar chart?

- a.** px.scatter()
- b.** px.line()
- c.** px.bar()
- d.** px.histogram()

60. Which keyword is used to define a function in Python?

- a.** func
- b.** define
- c.** def

d. function

61. A student writes the following Python code to check whether a number is even: `num = int(input("Enter a number: "))` if `num % 2 == 0`:
`print("Even")` else: `print("Odd")` However, the program gives incorrect results when the user enters a decimal value like 4.5. What is the best modification to make the program more robust?

- a.** Replace `int()` with `float()` and keep the same condition
- b.** Use `num // 2` instead
- c.** Use `num / 2 == 0` instead of modulus
- d.** Remove input and hardcode the value

62. Python is a:

- a.** Hardware
- b.** Programming Language
- c.** Operating System
- d.** Database

63. What will be the output of the following Python code? `x = 5` `y = 2`
`print(x > y and y > 1)`

- a.** True
- b.** False
- c.** Error
- d.** None

64. Which function is used to accept input from the user?

- a.** `read()`
- b.** `scan()`

c. `get()`

d. `input()`

65. What will be the output? `for i in range(3): print(i)`

- a.** 1 2 3
- b.** 0 1 2
- c.** 0 1 2 3
- d.** Error

66. Which data type is used to store whole numbers?

- a.** float
- b.** str
- c.** int
- d.** bool

67. What will be the output? `print("5" + "3")`

- a.** 8
- b.** 53
- c.** 15
- d.** Error

68. Which statement is used to display output in Python?

- a.** `input()`
- b.** `print()`
- c.** `display()`
- d.** `output()`

69. Which of the following is NOT a valid Python data type mentioned in the text?

- a.** int
- b.** float
- c.** character
- d.** bool

70. What type of values does the Boolean data type hold?

- a.** Whole numbers
- b.** Decimal numbers
- c.** Text
- d.** True or False

71. Which of the following is an example of a relational operator?

- a.** +
- b.** *
- c.** ==
- d.** and

72. Which logical operator returns True if both conditions are true?

- a.** or
- b.** not
- c.** and
- d.** !=

73. What is the purpose of the if statement in Python?

- a.** To repeat a block of code.
- b.** To define a function.

c. To execute a block of code only if a condition is true.

d. To store multiple items in a single variable.

74. Which conditional statement allows you to check multiple conditions in sequence?

- a.** if
- b.** if-else
- c.** nested if
- d.** if-elif-else

75. What is the term for repeating a block of code multiple times?

- a.** Selection
- b.** Iteration
- c.** Condition
- d.** Assignment

76. Which type of loop is used when you know the number of times you want to repeat a block of code?

- a.** while loop
- b.** for loop
- c.** if loop
- d.** nested loop

77. Which data structure is used to store multiple items in a single variable in Python, as shown in the example ["Computer", "Science", 20, True]?

- a.** String
- b.** Tuple
- c.** List

d. Dictionary

78. Which keyword is used to define a user-defined function in Python?

a. function

b. define

c. def

d. func

79. What are the values passed to a function when it is called known as?

a. Parameters

b. Arguments

c. Return values

d. Scope

80. The part of the program where a function can be accessed is called its:

a. Parameter

b. Argument

c. Return type

d. Scope

81. Variables declared inside a function have:

a. Global scope

b. Local scope

c. Unlimited scope

d. No scope

82. What does the return statement do in a Python function?

a. Prints output to the console.

b. Takes input from the user.

c. Sends a value back to the caller of the function.

d. Defines the function.

83. What are the values specified in the function header within the parentheses called?

a. Arguments

b. Parameters

c. Return values

d. Local variables

84. What type of argument allows you to call a function by specifying the parameter names?

a. Positional arguments

b. Default arguments

c. Keyword arguments

d. Required arguments

85. A function that performs an action but does not explicitly return a value is sometimes referred to as a:

a. Non-void function

b. Void function

c. Recursive function

d. Anonymous function

86. Can a Python function return multiple values?

a. No

b. Yes, as a list

- c. Yes, as a tuple
- d. Yes, directly separated by commas

87. A collection of modules that together supply specific needs or applications is called a:

- a. Module
- b. Package
- c. Library
- d. Function

88. Which of the following is a popular Python library used for data manipulation and analysis?

- a. Math
- b. Random
- c. Pandas
- d. Turtle

89. What is a container that contains various functions to perform specific tasks in Python?

- a. Module
- b. Library
- c. Package
- d. Function

90. Which statement is used to access modules or files from a Python package?

- a. include
- b. use

- c. import
- d. access

91. Which of the following is NOT a standard way to import a module in Python?

- a. import module_name
- b. from module_name get function_name
- c. import module_name as alias
- d. get function_name from module_name

92. Which Python library provides functions for generating random numbers?

- a. Math
- b. Random
- c. Pandas
- d. Matplotlib

93. What type of error occurs when the code violates the rules of the programming language's syntax?

- a. Runtime Error
- b. Logical Error
- c. Syntax Error
- d. Exception

94. Errors detected during program execution are called:

- a. Syntax Errors
- b. Logical Errors
- c. Exceptions

d. Warnings

95. Which block is used to test a block of code for potential errors in Python?

- a. catch**
- b. handle**
- c. try**
- d. error**

96. Which block in error handling always executes, regardless of whether an exception occurred or not?

- a. else**
- b. finally**
- c. except**
- d. try**

97. Which Python library is particularly useful for simplifying file handling of structured data like CSV files?

- a. math**
- b. random**
- c. pandas**
- d. turtle**

98. What is the primary function used in Pandas to read data from a CSV file?

- a. open()**
- b. read_file()**
- c. pd.read_csv()**

d. df.read_csv()

99. When reading a CSV file with Pandas, which parameter in `read_csv()` is used to specify the separator between values?

- a. separator**
- b. sep**
- c. delimiter**
- d. value_sep**

100. What attribute of a Pandas DataFrame can be used to obtain the header or field names after reading a CSV file?

- a. headers**
- b. columns**
- c. fields**
- d. names**

101. Which Pandas method is used to write a DataFrame to a CSV file?

- a. write_csv()**
- b. to_file()**
- c. df.to_csv()**
- d. pd.write_csv()**

102. When writing a DataFrame to a CSV file using `to_csv()`, what does the parameter `index=False` do?

- a. It includes the index as a column in the CSV.**
- b. It removes the header row from the CSV.**
- c. It excludes the index column from the CSV.**
- d. It sorts the data based on the index.**

103. What is the default separator used by Pandas when reading or writing CSV files?

- a. semicolon (;)
- b. tab (\t)
- c. comma (,)
- d. space ()

104. Which mode should be used with the built-in open() function if you want to read a file?

- a. "w"
- b. "a"
- c. "r"
- d. "x"

105. Which mode, when used with the built-in open() function, will overwrite the file if it exists or create a new file if it doesn't?

- a. "r"
- b. "a"
- c. "w"
- d. "x"

106. Which Pandas function is used to create a DataFrame from a dictionary or a list of lists?

- a. read_csv()
- b. to_csv()
- c. pd.DataFrame()
- d. create_df()

107. What is the primary goal of data visualization?

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- c. To understand data through visual context.
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- c.** Seaborn
- d.** GGPlot

c. px.bar()

d. px.histogram()

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- b.** plt.plot()
- c.** plt.bar()
- d.** plt.pie()

115. Which Plotly Express function is used to create a bar chart?

- a.** px.scatter()
- b.** px.line()

CHAPTER 5: ARTIFICIAL INTELLIGENCE (AI) AND CONTEMPORARY TECHNOLOGIES

1. An AI recruitment tool is trained on 10 years of company data where 90% of past successful engineers were male. The AI now automatically filters out female resumes. What is the fundamental issue here, and how should it be solved?

- a. Cloud latency; upgrade to 5G.
- b. Algorithmic bias; the training data must be diversified to be fair and representative.
- c. Unsupervised learning; it should be changed to robotics.
- d. Data privacy; the resumes should be encrypted.

2. If a national government moves all its critical citizen data (like tax and health records) to a public cloud service, what is the most significant risk they face?

- a. The data will consume too much electricity.
- b. Citizens will not be able to use e-Commerce.
- c. Severe security and data breach risk if the cloud provider suffers a cyber-attack or data breach.
- d. The cloud will convert the data into Virtual Reality.

3. How does the rise of Generative AI tools (like ChatGPT) severely complicate the concept of "Academic Honesty" in schools?

- a. It prevents students from submitting handwritten assignments.
- b. It allows students to submit generated essays as their own original work, making plagiarism difficult to detect.

c. It requires all students to have laptops.

d. It forces teachers to use VR headsets.

4. In developing a "Smart City," how must IoT and Cloud Computing integrate to function effectively?

- a. Cloud computing generates the electricity that powers the IoT sensors.
- b. IoT sensors collect real-time city data (traffic, pollution), which is sent to the Cloud for massive processing, storage, and AI analysis to optimize city services.
- c. They only communicate to monitor the weather.
- d. They do not integrate; they are competing technologies.

5. What is the primary societal risk of mandatory e-Education in a developing country with diverse geographical regions?

- a. Students will become too skilled.
- b. The "Digital Divide" - a lack of reliable internet access, devices, or digital literacy, leading to educational inequality.
- c. e-Education requires students to buy VR headsets.
- d. Teachers cannot learn how to use SaaS.

6. A self-driving car (an AI/ML application) is involved in a fatal accident. What is the primary ethical and legal dilemma presented by this scenario?

- a. The car's paint color was wrong.
- b. Determining who is at fault: the human driver, the AI software developer, or the car manufacturer.
- c. The song did not stop playing.
- d. The car's IoT sensors were not connected to a smartphone.

- 7.** Why would a large hospital utilize a hybrid cloud approach (using both private and public clouds) instead of exclusively using a public cloud?
- a.** To keep sensitive patient data secure on a private cloud while using the public cloud for scalable, less-sensitive operations.
 - b.** Because public clouds cannot run AI software.
 - c.** To ensure that their IoT devices can only use supervised learning.
 - d.** Because private clouds are cheaper to maintain than public ones.
- 8.** Analyze the use of unsupervised learning in medical diagnostics. How might it potentially discover new disease patterns?
- a.** By relying on doctors to manually label every case.
 - b.** By analyzing massive amounts of raw, unlabelled patient data to find hidden clusters of symptoms that humans haven't previously categorized together.
 - c.** By using Gen AI to write fake medical journals.
 - d.** By filtering spam emails from hospital servers.
- 9.** Evaluate the psychological impact of Augmented Reality (AR) in technical training compared to reading a manual. Why is AR superior?
- a.** AR replaces the real world entirely, eliminating distractions.
 - b.** AR overlays interactive, step-by-step digital schematics directly onto physical machinery in real-time, drastically reducing cognitive load and errors.
 - c.** AR is cheaper to print than paper manuals.
 - d.** AR uses unsupervised learning to read the manual aloud.

- 10.** How does the integration of AI in cybersecurity shift the defense paradigm from "reactive" to "proactive"?
- a.** AI completely unplugs the network when attacked.
 - b.** AI waits for a virus to destroy files before sending an email alert.
 - c.** AI analyzes network traffic patterns continuously to predict and neutralize threats before they can execute, rather than just cleaning up after an attack.
 - d.** AI replaces human security guards with physical robots.
- 11.** Which type of AI can create images, text, and music?
- a.** Narrow AI
 - b.** Generative AI
 - c.** Reactive AI
 - d.** Traditional AI
- 12.** Which of the following is a risk of AI?
- a.** Faster processing
 - b.** Better automation
 - c.** Privacy and security concerns
 - d.** Improved accuracy
- 13.** Which of the following is an application of AI?
- a.** Smart Assistants
 - b.** Face Recognition
 - c.** Chatbots
 - d.** All of the above

14. Which of the following is a service of cloud computing?

- a.** IaaS
- b.** PaaS
- c.** SaaS
- d.** All of the above

15. Which of the following is not related to AI?

- a.** Behaving like human
- b.** Learning from past experience
- c.** Putting human brain in machine
- d.** Solving problems themselves

16. Which of the following is an example of AI?

- a.** Self-driving cars
- b.** Siri
- c.** Alexa
- d.** All of the above

17. What is Machine Learning?

- a.** Programming without computers
- b.** A method that enables computers to learn from data
- c.** Internet browsing
- d.** Data storage

18. What is the main purpose of cybersecurity?

- a.** Entertainment
- b.** Protecting data and systems
- c.** Gaming

d. Networking only

19. What is Big Data?

- a.** Small amount of information
- b.** Extremely large and complex datasets
- c.** Printed reports
- d.** Computer hardware

20. What is the primary goal of contemporary technologies?

- a.** Increase manual work
- b.** Improve efficiency and solve real-world problems
- c.** Reduce communication
- d.** Eliminate computers

21. Which of the following best defines Artificial Intelligence (AI)?

- a.** Making machines faster than humans
- b.** Teaching machines to follow commands exactly as programmed
- c.** Enabling machines to think and act like humans
- d.** Using only mathematical formulas in computers

22. Which field helps AI systems understand and use human languages?

- a.** Biology
- b.** Linguistics
- c.** Psychology
- d.** Physics

23. In which type of Machine Learning is the computer trained with labeled data?

- a.** Unsupervised Learning
- b.** Supervised Learning
- c.** Reinforcement Learning
- d.** Semi-supervised Learning

24. What makes a robot "smart" or intelligent?

- a.** Adding more sensors only
- b.** Giving it a mechanical body
- c.** Programming it with fixed rules only
- d.** Using AI to make decisions and adapt to situations

25. Which of the following is a limitation of AI tools?

- a.** They save time and encourage creativity
- b.** They can work without internet access
- c.** They may sometimes produce incorrect or incomplete information
- d.** They always provide accurate and verified answers

26. Which feature of Google Docs helps users complete sentences while typing?

- a.** Smart Compose
- b.** Voice Typing
- c.** Grammar Check
- d.** Help Me Write

27. In Microsoft PowerPoint, which AI feature gives real-time feedback on speaking pace and tone?

- a.** Designer
- b.** Translator
- c.** Presenter Coach
- d.** Live Captions

28. Which of the following is an example of AI bias?

- a.** A chatbot explaining science topics
- b.** A job recruitment system that favors one gender
- c.** A translation tool converting languages
- d.** A navigation app showing the shortest route

29. What is the best way to protect personal data while using AI tools?

- a.** Use strong passwords and avoid sharing sensitive information
- b.** Post private photos publicly
- c.** Share all personal details for better accuracy
- d.** Accept all requests from unknown emails

30. A home thermostat that senses room temperature, sends data to the cloud, and auto-adjusts the AC illustrates which IoT cycle?

- a.** Sense -> Send -> Process -> Act
- b.** Store -> Execute -> Report
- c.** Design -> Build -> Test
- d.** Input -> Output only

31. XR (Extended Reality) is best described as:

- a.** Only VR headsets

b. A sub-feature of gaming graphics

c. A video editing technique

d. An umbrella term covering VR, AR, and MR

32. Which cloud service model lets users run ready-to-use apps via a browser, without installing them?

a. IaaS

b. PaaS

c. SaaS

d. DBaaS

33. A school stores public brochures on Google Drive but keeps confidential student records on its own servers. This is best described as:

a. Public Cloud

b. Hybrid Cloud

c. Private Cloud

d. Community Cloud

34. Selling used items on hamrobazar.com by one person to another is which e-Commerce model?

a. B2C

b. B2B

c. C2C

d. C2B

35. Which component of e-Governance focuses on services like online passports and tax payments for citizens?

a. G2C

b. B2C

c. G2G

d. C2G

36. Which of the following is not related to AI?

a. Behaving like human

b. Learning from past experience

c. Putting human brain in machine

d. Solving problems themselves

37. Which of the following is an example of AI?

a. Self driving cars

b. Siri

c. Alexa

d. All of the above

38. Which of the following is not an application of AI?

a. Gaming

b. Automated systems

c. Expert systems

d. Manual writing

39. Which AI chatbot was developed by Google?

a. DeepSeek

b. Copilot

c. ChatGPT

d. Gemini

40. Which of the following is a service of cloud computing?

- a.** IaaS
- b.** PaaS
- c.** SaaS
- d.** All of the above

41. A hospital uses a computer system that analyses X-ray images and suggests possible diseases. Which technology is being used?

- a.** Word processing
- b.** Artificial Intelligence
- c.** Spreadsheet
- d.** Database

42. Which practice best helps reduce bias in AI systems?

- a.** Using only one type of data
- b.** Ignoring privacy rules
- c.** Using diverse and representative training data
- d.** Copying outputs without verification

43. Which situation best demonstrates Machine Learning?

- a.** A calculator performing arithmetic operations
- b.** A computer learning to identify spam emails from previous examples
- c.** A printer producing documents
- d.** A monitor displaying images

44. An autonomous vehicle identifies road signs and obstacles while driving. Which AI capability is primarily involved?

- a.** Computer Vision

b. Word Processing

c. Database Querying

d. Networking

45. Which of the following is not related to AI?

- a.** Behaving like human
- b.** Learning from past experience
- c.** Putting human brain in machine
- d.** Solving problems themselves

46. Which of the following is an example of AI?

- a.** Self driving cars
- b.** Siri
- c.** Alexa
- d.** All of the above